# Scalability & Future Plan

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| **Date** | 30 October 2026 |
| **Team ID** | Individual |
| **Project Name** | EV Market Analysis |
| **Maximum Marks** | 1 Mark |

### Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

### Current System Limitations:

| S.No | Limitation | Impact | Priority to Address (High / Medium / Low) |
|---|---|---|---|
| 1 | Dashboards use a static dataset with no live data refresh | Insights become outdated as new EV sales data arrives | High |
| 2 | Heavy charts (scatter, map) with many manufacturers | Slower loading and cluttered visuals with larger data | Medium |
| 3 | No role-based access or user-level views | All viewers see the same data and filters | Low |

### Scalability Plan:

| S.No | Scalability Aspect | Current State | Proposed Upgrade / Solution |
|---|---|---|---|
| 1 | User Load | Single user on Tableau Desktop | Publish to Tableau Cloud / Server to support many concurrent viewers |
| 2 | Data Storage | Local data file / extract | Move data to a cloud database (e.g., SQL, BigQuery) with scheduled refresh |
| 3 | Performance | Works well on current data size | Use extracts, data aggregation and fewer filters to keep dashboards fast |
| 4 | Security | No access restrictions | Add row-level security and role-based access on the server |

### Future Roadmap:

| Phase | Planned Feature / Enhancement | Target Timeline | Expected Impact |
|---|---|---|---|

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|---|---|---|---|
| Phase 2 | Live data refresh from a cloud database | 1-2 months | Always up-to-date insights |
| Phase 3 | Publish dashboards online with role-based access | 3-4 months | Wider access for stakeholders |
| Phase 4 | Add predictive analytics (sales forecasting) | 5-6 months | Better planning decisions |